

Overview

This report presents a restaurant analysis using Zomato datasets. Zomato is a multinational restaurant aggregator and food delivery platform. The objective of this analysis is to address the following key questions:

1. Which restaurants are the most popular?
2. Which restaurants generate the highest total sales?
3. What is the average meal price for the top 10 restaurants by total sales?
4. How do total sales trend over time?
5. Which cities generate the highest total sales?
6. Which type of food has the higher average price: vegetarian or non-vegetarian?

Hypothesis

The hypothesis is that meal price is a key factor influencing total sales. In general, customers tend to choose meals that are reasonably priced, neither too high nor too low, as they seek both affordability and assurance of quality.

Methodology

The Zomato dataset consists of five tables, four of which were used in this analysis: Orders, Food, Menu, and Restaurants. Data cleaning involved removing null values and eliminating duplicate entries in each table's primary key. The Orders table was joined with the Restaurants table (many-to-one), as each restaurant can have multiple orders. Similarly, the Menu table was joined with the Food table (many-to-one).

A new column was added to the Orders table using an IF statement: if the currency of the sales value was not in USD, it was converted from Indian Rupees to USD by multiplying by 0.012. The same conversion was applied to the price column in the Menu table. Finally, nine different charts were created to answer the key analysis questions.

Visualizations

1. **Line Chart** – Total Sales by Year and Month (shows the sales trend over time).
2. **Column Chart** –
 - Top 10 Restaurants by Total Sales (compares total sales across top restaurants).
 - Average Meal Price by Restaurant (compares average meal prices across restaurants).
3. **Card / KPI Tiles** –
 - Total Sales
 - Total Sales Quantity
 - Average Meal Price
4. **Table** – Most Popular Restaurants (restaurant name with rating count).
5. **Bar Chart**– Total Sales by City (compares total sales across cities).
6. **Pie Chart** – Average Price Distribution: Veg vs Non-Veg (compares meal prices by type).

Data Modeling

